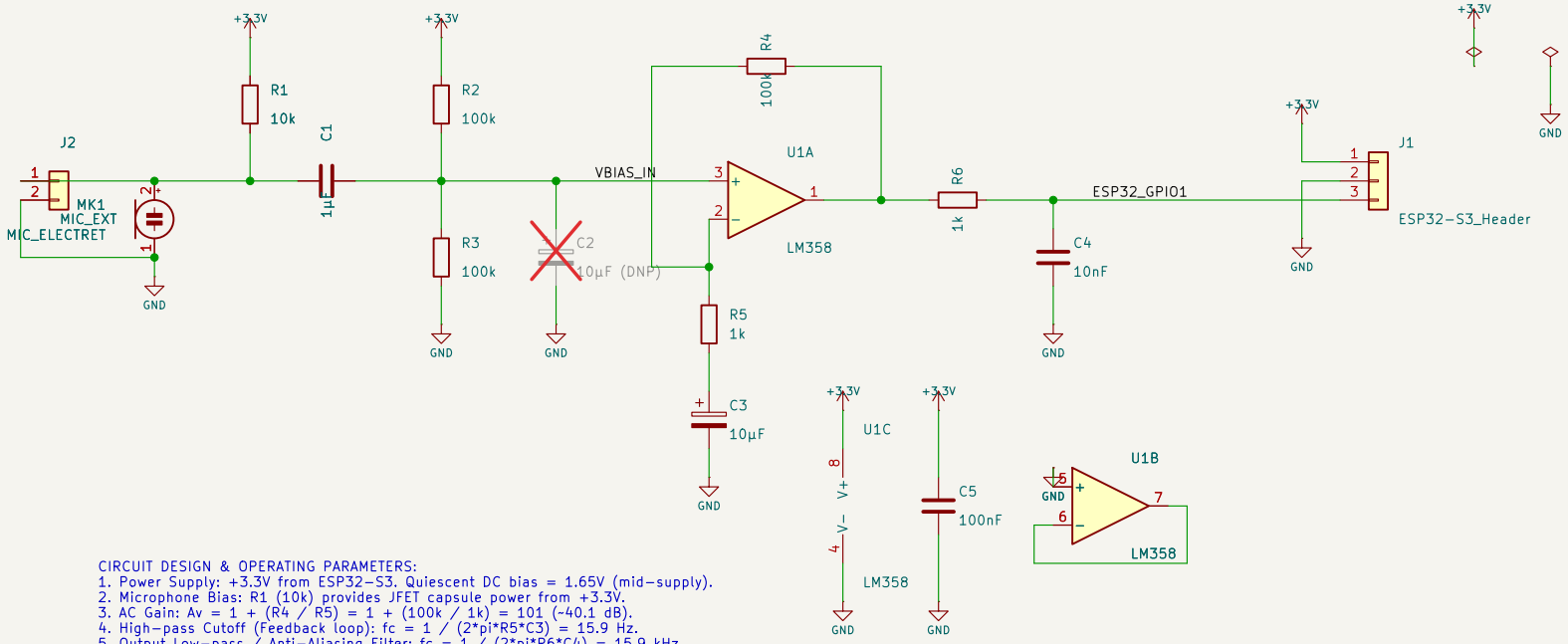


CIRCUIT DESIGN & OPERATING PARAMETERS:
1. Power Supply: +3.3V from ESP32-S3. Quiescent DC bias = 1.65V (mid-supply).
2. Microphone Bias: R1 (10k) provides JFET capsule power from +3.3V.
3. AC Gain: Av = 1 + (R4 / R5) = 1 + (100k / 1k) = 101 (~40.1 dB).
4. High-pass Cutoff (Feedback loop): fc = 1 / (2*pi*R5*C3) = 15.9 Hz.
5. Output Low-pass / Anti-Aliasing Filter: fc = 1 / (2*pi*R6*C4) = 15.9 kHz.
6. NOTE ON C2 (10uF): In the original prompt ASCII, C2 was drawn across R3 at Pin 3. If populated directly on Pin 3, C2 shorts out audio signals (>30Hz). Marked DNP to preserve audio bandwidth. For filtered Vbias, add a series isolation resistor.

Target Microcontroller: ESP32-S3 GPIO 1 (ADC1_CH0)
Gain: 40 dB (x101), Low-pass Cutoff: 15.9 kHz
Antigravity Hardware Design

Sheet: /
File: esp32s3_mic_preamp.kicad_sch

Title: ESP32-S3 Electret Microphone Preamplifier (LM358)

Size: A4	Date: 2026-10-03	**Rev: v1.0**
KiCad E.D.A. 10.0.6		Id: 1/1